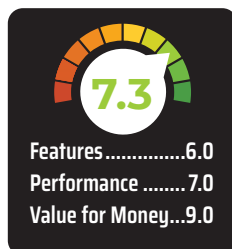


AMD Ryzen 7 7700X3D

Almost as fast, a lot cheaper... if AMD prices it right

It's been three years since AMD's Ryzen 7 7800X3D established a straightforward formula for gaming-focused desktop processors by combining eight Zen 4 cores with a large pool of vertically stacked cache along with slightly lower clock speeds in exchange for stronger performance in latency-sensitive games. Knock off a few multipliers off the base and boost clocks, and you've got the Ryzen 7 7700X3D which follows that same basic template. It goes without saying that the 7700X3D does not feature a fundamentally new processor design. Like the 7800X3D, the Ryzen 7 7700X3D uses a single eight-core Zen 4 compute die equipped with first-generation AMD 3D V-Cache. It consequently offers eight cores, 16 threads and 96 MB of L3 cache. The principal difference, like we mentioned, is with the frequencies. Its 4.0 GHz base clock is 200 MHz below that of the 7800X3D, while its maximum boost clock falls from 5.0 GHz to 4.5 GHz.

The resulting product is positioned as a more accessible route into AMD's cache-heavy gaming line-up. It also gives existing AM5 owners another potential drop-in upgrade, subject to motherboard BIOS support. AMD describes the processor as a way to broaden the choices available to builders upgrading within AM5 or entering the platform for the first time. The Ryzen 7 7700X3D needs to provide a meaningful gaming advantage over the ordinary Ryzen 7 7700X, while remaining sufficiently differentiated from the faster 7800X3D.



TEST RIG

We've tested the Ryzen 7 7700X3D with the NVIDIA GeForce RTX 5090 as well as the Radeon RX 9070 to check out the Rebar performance impact. Aside from that, the platform used was the Colorful iGAME X870E

Vulcan OC 14, 32 GB of G.Skill Trident Z5 Neo DDR5-6000 memory, and Samsung's 9100 Pro NVMe SSD. Cooling is handled by the Corsair H150i, while a Seasonic Focus Gold 850W PSU powers the system.

PERFORMANCE

Cinebench 24

Cinebench 24 is based on Maxon's Cinema 4D software, designed to evaluate a processor's performance in rendering complex 3D scenes. It tests both single-core and multi-core capabilities, highlighting how efficiently a CPU handles multi-threaded tasks common in professional rendering workflows. The benchmark utilizes modern instruction

sets and large datasets, providing an up-to-date assessment of CPU performance in contemporary applications.

Blender

The Blender Benchmark assesses a processor's performance by rendering scenes. It evaluates both CPU and GPU capabilities in handling complex rendering tasks. This benchmark provides insights into how well a processor performs in real-world 3D modeling, animation, and rendering scenarios. An interesting aspect is that it highlights the efficiency of different CPU architectures in multi-threaded workloads, as Blender can utilize multiple cores. It's particularly useful for professionals and enthusiasts relying on Blender, indicating how hardware upgrades might impact workflow efficiency and rendering times.

V-Ray

V-Ray is another benchmark that measures a processor's performance in rendering but it uses the V-Ray engine, widely used in visual effects, architecture, and design industries. It tests both CPU and GPU rendering capabilities, focusing on multi-threaded performance and handling complex calculations involved in ray tracing. CPUs with higher core counts and multi-threading technology tend to excel here as V-Ray efficiently utilises available threads.

Procyon Office

This benchmark measures a processor's performance in real office applications like Microsoft Office suite, testing tasks such as document editing, spreadsheet calculations, and presentation creation. It focuses on both single-threaded and multi-threaded performance, providing insights into how a CPU handles everyday productivity tasks.

7 Game Average

While 3DMark is an excellent synthetic benchmark, it's not indicative of real-world performance and that's why